

Sodium-ion in the field: eighteen months of grid storage data

Gridfield Analytics, field report GF-2026-09, August 2026. Independent; no vendor funding.

Summary

We collected operating data from 312 grid storage sites in three climate zones that installed sodium-ion systems between 2024 and 2025. Projected cycle life to 80 percent capacity is 2,900 cycles, well below the 4,000-plus cycles reported by laboratory studies.

1. Why field numbers are lower

Field sites cycle at partial state of charge, see temperatures from -15 °C to 41 °C and run 1.4 cycles a day on average. Sites in the cold zone degraded fastest: 2,300 cycles projected.

2. Availability

Fleet availability was 97.1 percent. Most outages were inverter faults, not cell failures.

3. Implications for models

Analysts who use laboratory cycle-life figures will overstate lifetime energy throughput by about 30 percent and understate levelised cost of storage accordingly.